

orbit_demo design overview

Package section 01_overview of orbit_demo_design_review, written 2026-10-06 from the package files and the repository at commit 3dffffbb. Links are relative to this file. Cited reports/... files are linked to their byte-identical copies in 07_verification. Unlinked repository paths under rtl/, pd/, tb/, fault/ and scripts/ are copied in 04_digital_design/source/. Only build/... files without a package copy (the copies are listed in README.md under How the package was built), docs/research/... and ORFS image paths (platforms/..., flow/scripts) are outside the package. Discrepancies are listed in the discrepancy register (09_review_notes).

Status labels: **VERIFIED** = a tool log or report shows it. **REPRODUCED 2026-10-06** = re-run for this package. **TARGET** = design goal, not measured. **ASSUMPTION. CLAIM (unverified)** = stated in a document, no log found. **MISSING. N/A** = not applicable (reason given).

Design revision covered by this package

Item	Revision / identifier	Note
Design	orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)	top module orbit_demo, LANES=4
RTL	rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26d71a8 (work dated 2026-09-29 18:54 UTC; see the note on commit dates below); unchanged through the package commit	sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256 at the package root)
Specification	docs/SPEC.md at 26d71a8 (never revised)	sha256 aef5d9fb...
Concept brief	docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)	sha256 9d5f33d3...
Reviewed layout	ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/	6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spef 014e655a...
Layout configuration	pd/sky130hd_sep/config.mk and regions.tcl at 5e54769; constraint.sdc at 3e8dad1 (set clk_period 7.2)	The run (21:56 UTC) used config.mk/regions.tcl as in d8ead98, identical to 5e54769 once comments are stripped. It used the working-tree constraint.sdc with period 7.2 ns, which is the version in commit 3e8dad1 (committed at 22:00 UTC, after the run had started) and is byte-identical to the flow copy build/pd_sep/sdc/sky130hd/sep.sdc (written 21:56 UTC). d8ead98's constraint.sdc has a 7.0 ns period, so a rebuild must not use it
Flow / tools	ORFS docker image tag openroad/orfs:latest; the only local image is sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC, before the run). Linking the run to this digest is an ASSUMPTION: the run logged no digest. OpenROAD prints version "unknown". KLayout 0.30.12 (DRC/LVS). Yosys 0.69+154 (local sims/schematics)	No PDK commit is recorded in the logs
Process / library	SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)	—
Package assembled	2026-10-06; design inputs copied from the repository tree at commit 3dffffbb (branch claudе/hopeful-rubin-0yf8io)	Every later commit on the branch changes only review/ or the repository landing page README.md (added after packaging; not a design input), or is empty (git diff --name-only 3dffffbb..HEAD outside review/ lists only README.md). Re-runs made for this package are dated 2026-10-06

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout.

Note on commit dates: the commit identifiers above are on the current branch and have the same file trees as the commits the package was built from ([COMMIT_MAP.tsv](#) lists them with tree hashes). All dates in this package come from the tool logs and run records, which are dated 2026-09-29 to 2026-10-06. The author and committer dates in the branch's commit metadata do not match these and should not be read as the development timeline (D-54).

Notes on this table, checked for this document:

- MANIFEST.sha256, cited in the RTL row, is at the package root: [MANIFEST.sha256](#), generated last on 2026-10-06 (see [README.md](#) section Manifest; discrepancy register (09_review_notes) D-53). The four RTL sha256 values recomputed at 3dffffbb equal those in [fault/summary.md](#) lines 16-19 and in [04_digital_design/README.md](#); e.g. rtl/orbit_demo.v
a3ffffb229df3e166694d9259cee45411537118aa2673f84bb26b0f4108f3183a.

Purpose and scope

orbit_demo is the "BUILT / DIGITAL DEMONSTRATOR" box of the ORBIT-AI v0.1 concept brief, not the 16-tile chip ([SPEC.md](#) lines 3-7; [brief](#) p.1). It is a standard-cell digital block: four independent signed INT8 multiply / INT32 accumulate lanes behind a valid/ready input stream, a one-result output buffer, duplicated accumulator and result storage with a sticky mismatch fault-stop, and a three-copy (TMR) thermal state machine that throttles, stops and recovers input admission from a digital temperature input ([orbit_demo.v](#) lines 1-16). It was taken through OpenROAD-flow-scripts on sky130hd to a routed GDS and was not fabricated (brief p.1 "No physical chip fabricated."; `finish_design_instance_count_padcells 0` in [6_report.json](#) line 60). There are no analog blocks, no pad ring and no package. [SPEC.md](#) lines 9-10: "Nothing here is radiation-qualified, timing-signed-off for flight or thermally validated. Thresholds are illustrative."

Item	Demonstrator orbit_demo (built, reviewed here)	ORBIT-AI concept chip (brief only, not built)
Compute	4 lanes x 1 INT8 MAC, INT32 accumulate	16 tiles x 32 x 32 = 16384 INT8 MACs: TARGET
Clock	no requirement (MISSING); 7.2 ns met at TT (ORFS optimised at TT only): VERIFIED; re-time of the same layout passes at ff_n40C_1v95 and fails setup at ss_100C_1v60 (WNS -6.131 ns): REPRODUCED 2026-10-06	200-800 MHz, 6.55-26.21 TOPS: TARGET ("No timing measurement supports these clock targets", p.2)
Storage	521 flip-flops; no SRAM or ECC (rtl/ecc/ exists but is not instantiated: N/A here)	32 MiB SRAM with SECCDED 64+8: TARGET
Protection	duplicate + compare fault-stop (512 bits), TMR + repair (6 bits), 3 unprotected flip-flops, no replay	duplication, checkpoint/retry, tile quarantine: TARGET
Power	no budget (MISSING); 52.1092 mW default-activity estimate: VERIFIED tool output	40 W "for conceptual sizing only": ASSUMPTION
Process	SKY130 / sky130_fd_sc_hd via ORFS: VERIFIED	"process, package ... remain undecided" (p.2)

Requirements and verification status

Evidence keys (location and test conditions):

Key	Evidence	Conditions	Status
E-FV	formal_summary.md	make formal, 12 Symbyosys tasks: k-induction k = 1-8, abc pdr, suprove liveness, cover depth 24/8; Yosys 0.69+154, SBY v0.69, Yices 2.7.0, bitwuzla 0.9.1; reference model, inputs free after reset in cycle 0. datapath_pdr re-run 2026-10-06: PASS, 941 s (make_formal-extra.log , summary_extra.md); published result in formal/summary.md	REPRODUCED 2026-10-06

Key	Evidence	Conditions	Status
E-VAC	formal_vacuity.md	19 one-line RTL mutants, 19 CAUGHT	REPRODUCED 2026-10-06
E-SIM	directed_icarus.log , directed_verilator.log	tb/tb_orbit_demo.v; Icarus 14.0 (devel) and Verilator 5.053; seed 1; zero-delay RTL, 10 ns sim clock; 14 scenarios, 133601 cycles, 0 errors	REPRODUCED 2026-10-06
E-RND	random_icarus_summary.md	cocotb; seeds 1-5 x 20000 cycles + wrap_soak (149226 cycles); 7544 results, 0 mismatches	REPRODUCED 2026-10-06
E-WAV	waveforms/README.txt , negctl_summary.txt	5 self-checking VCD scenarios (Icarus 14.0, RTL zero-delay), all PASS; each FAILs on an RTL copy with its targeted bug	REPRODUCED 2026-10-06
E-GLSF	gls.log	pd/gls/tb_gls.v: RTL and routed sep 6_final.v in lockstep, zero-delay, 30000 cycles, seed 1, one flip per 32 cycles into the 518 redundant bits; run 2026-09-29	VERIFIED
E-FI	fault/summary.md , fault/logs/ ; re-run formal_results.md , make_fault.log	generated RTL copies, single upset per trace (any lane/bit/cycle); prove k-induction $k = 4$, negatives BMC depth 12; SBY v0.69, Yices 2.7.0. Re-run 2026-10-06 (19 tasks, 0 unexpected), in 07_verification/rerun_2026-10-06/fault/	REPRODUCED 2026-10-06
E-SEU	fault/campaign.md ; re-run campaign_report.md (byte-identical)	fault/tb_seu_campaign.v, Icarus 14.0, seed=1 tpb=12 tpb_unprot=200, 6836 trials, flips at the falling edge	REPRODUCED 2026-10-06
E-AUD	storage_audit.txt ; synth/storage_audit.txt	pd/audit_storage.py on sep 6_final.v (18/18 with the older script; current script 19/19, re-run 2026-10-06: storage_audit_current_script.txt); scripts/storage_audit.py on the generic Yosys 0.69+154 netlist	VERIFIED
E-MUT	mutation/summary.md	40 RTL mutants x 5 make targets; evidence bench tb_mutation_gaps.v	VERIFIED
E-PIN	pinout.csv , rtl_ports.json	parse_pins.py on sep 6_final.def; Yosys 0.69+154 elaboration	REPRODUCED 2026-10-06
E-STA	sta_corners_sep_2026-10-06/	pd/sta_corners.tcl on sep 6_final.{odb,sdc,spef}, 7.2 ns, nominal OpenRCX SPEF at every corner	REPRODUCED 2026-10-06

SPEC sections 2-9 (the §9 claim number is given in brackets), then the brief's claims and targets:

Requirement	Source	Status	Evidence
R2 18 ports, names/directions/widths per §2 (80 in, 135 out bits)	SPEC 25-45	REPRODUCED 2026-10-06	E-PIN: RTL and DEF both 18 ports / 215 signal bits, 0 mismatches (see Interfaces)
R3.1 exact signed 8 x 8 product, sign-extended to 32 bits [9.1]	SPEC 52-57	REPRODUCED 2026-10-06	E-FV P1 (product $k = 1$ over all 2^{16} operand pairs; datapath $k = 6$); E-SIM signed_edges; E-VAC zext_product
R3.2 accumulate mod 2^{32} , no saturation or overflow flag [9.5]	SPEC 56	REPRODUCED 2026-10-06	E-FV P1, wrap covers C20-C22; E-SIM int32_wrap; E-RND wrap_soak
R3.3 in_last -> result, out_valid next cycle; in_first && in_last legal; sums continue across in_last; lanes independent	SPEC 59-64	REPRODUCED 2026-10-06	E-FV P1, P2; E-SIM first_and_last, continue_after_last; E-WAV a_mac3
R4.1 §4 out_valid/in_ready equations; full buffer blocks every beat [9.2]	SPEC 68-74	REPRODUCED 2026-10-06	RTL orbit_demo.v 64, 70-71 identical to SPEC; E-FV P2, P4; E-VAC no_backpressure; E-SIM backpressure; E-WAV b_backpressure
R4.2 in_last with out_fire in one cycle replaces the result, no bubble	SPEC 75-76	REPRODUCED 2026-10-06	E-FV P2 (incl. live); E-SIM back_to_back
R4.3 out_data stable while out_valid && !out_ready (no-fault case)	SPEC 77	REPRODUCED 2026-10-06	E-FV P3. SPEC omits that reset/clear_fault withdraw a presented result (see discrepancy register (09_review_notes))

Requirement	Source	Status	Evidence
R4.4 out_ready -> in_ready combinational	SPEC 78	REPRODUCED 2026-10-06	RTL 70-71; timed in-to-out, TT slack 2.577 ns (supplement). clear_fault -> in_ready is also combinational and undocumented (see discrepancy register (09_review_notes))
R5.1 two copies each of acc and result per lane, own feedback, shared product; out_data from copy A	SPEC 82-84	VERIFIED	RTL orbit_mac_lane.v 25-60; E-AUD. Copy-A source checked only by tb_mutation_gaps.v T2 (E-MUT: mutant m07 survives all make targets)
R5.2 mismatch blocks out_valid/in_ready in the same cycle; fault from the next cycle [9.3]	SPEC 86-92	REPRODUCED 2026-10-06	E-FI acc_a/acc_b/res_a/res_b prove PASS; E-SEU 6136 of 6136 detected at latency 1; E-WAV c_fault
R5.3 fault sticky until clear/reset	SPEC 88, 92	REPRODUCED 2026-10-06	E-WAV c2_fault_sticky and its negative control. No make target checks it (E-MUT: m22 fault_not_sticky survives)
R5.4 one acc/result upset never transfers a wrong out_data; it stops	SPEC 93-94	REPRODUCED 2026-10-06	E-FI D_out_fire_matches_ref proven; E-SEU 0 escapes in 6144 trials; E-GLSF 702 of 702 flips stopped on the routed netlist
R5.5 clear_fault/reset clear fault_q, out_valid_q, zero both copies, keep thermal state; no replay	SPEC 95-98	REPRODUCED 2026-10-06	E-FV P8 (clear from an unconstrained state); E-VAC clear_keeps_fault, clear_keeps_acc; E-SIM clear_fault, reset_midrun
R6.1 copies u_thermal.u_copy0/1/2, vote, rewrite every cycle, reset STOP; \$6 next-state table, signed t [9.4]	SPEC 102-116	REPRODUCED 2026-10-06	E-FV P5; E-VAC threshold mutants; E-SIM throttle, stop_and_recover, negative_temps; E-WAV d_thermal (383 checks)
R6.2 admission: NORMAL always, THROTTLE alternate (first admits), STOP never; drain in STOP	SPEC 118-124	REPRODUCED 2026-10-06	E-FV P6, P4; E-SIM throttle, drain_in_stop. SPEC phase wording differs from orbit_thermal_tmr.v line 77 (see discrepancy register (09_review_notes))
R6.3 voted code 3 behaves as STOP	SPEC 43, 115, 124	VERIFIED	only tb_mutation_gaps.v T3 (E-MUT); formal code-3 branches are vacuous; mutant m27 survives every make target
R6.4 a single upset thermal copy is outvoted and repaired; therm_repair	SPEC 126-127	REPRODUCED 2026-10-06	E-FI therm_c0/c1/c2 PASS; E-SEU 72 of 72 REPAIRED; E-GLSF 234 of 234 repaired
R7 521 flip-flop bits, 518 in 9 groups, 3 unprotected; synthesis keeps copies distinct [9.6]	SPEC 129-145, 174	VERIFIED	E-AUD: routed 521 sky130_fd_sc_hd_dfxt1, "RESULT: PASS (18/18 checks passed)" (19/19 with the current script); generic 9/9; without keep_hierarchy copies merge (521 -> 517 bits, audit FAIL)
R8.1 measure out_valid_q escapes	SPEC 156	REPRODUCED 2026-10-06	E-SEU 200 of 200 escape (38 LOST, 162 EXTRA); E-FI neg_out_valid_q FAIL as expected
R8.2 measure fault_q 1->0	SPEC 157-158	MISSING	single-upset flows cannot reach it; needed: a double-upset scenario
R8.3 measure phase	SPEC 159	REPRODUCED 2026-10-06	E-SEU 156 MASKED, 44 TIMING; shifted vs doubled admission not separated
R8.4 measure shared multiplier	SPEC 160	REPRODUCED 2026-10-06	E-FI neg_product: wrong out_data with fault = 0
R8.5 measure comparators, voter, clock, reset, sensor/handshake inputs, common-mode	SPEC 160-162	MISSING	not injected in any flow; needed: transient injection on vote/next-state, in_fire, clr, mismatch, clock, reset
R9.7 faults only in generated copies; no fault ports in production RTL	SPEC 175-176	REPRODUCED 2026-10-06	E-PIN: 18 functional ports, no test port
B1 "521 flip-flop bits"; "Nine storage groups audited"	brief p.4	VERIFIED	R7

Requirement	Source	Status	Evidence
B2 "4,356 generic logic cells"	brief p.4	CLAIM (unverified)	not reproduced: Yosys 0.69+154 gives 4373, YoWASP 0.69 gives 4384 (synth/summary.md lines 12, 21); no script given (see discrepancy register (09_review_notes))
B3 bounded results: "24 bounded SAT time steps", fault "8 time steps", wrap "6 time steps"	brief p.4	CLAIM (unverified)	no script or log; the repo's unbounded proofs (R3-R6) cover the behaviour
B4 "No PDK mapping or physical area estimate"; "OpenROAD was not run"	brief p.4-5	N/A	earlier project state, superseded by this layout (see discrepancy register (09_review_notes))
B5 thresholds 80 / 95 / 70 C, "not device ratings"	brief p.3	TARGET	RTL parameters orbit_demo.v 22-24; behaviour in R6.1
B6 "Separate redundant state physically"	brief p.3	VERIFIED	separation.md : acc/res same-bit centre min 251.17 um, copy A-B flip-flop gap 242.94 um, thermal 97.92 um / gap 92.54 um, 0 of 262 pairs touching. Geometry only. The 20 um / 10 um acceptance limits are a TARGET with no stated rationale
B7 "protect clock/reset distribution"	brief p.3	MISSING	not implemented: one CTS tree for all 521 sinks (4_1_cts.log line 17)
B8 16 tiles x 32 x 32 MACs, 6.55-26.21 TOPS at 200-800 MHz	brief p.1-2	TARGET	concept chip, not built
B9 40 W per chip	brief p.1-2	ASSUMPTION	"not a power estimate" (p.2)
B10 32 MiB SECDED SRAM; 16 GiB+ external memory	brief p.1-2	TARGET	concept chip; N/A for orbit_demo (no SRAM)
B11 "No physical chip fabricated"; sensors, calibration, shutdown circuit not implemented	brief p.1, p.3	N/A	scope statements; consistent with padcells 0 and the digital temp_c input

Manufacturing process

Item	Value	Source	Status
Process	SkyWater SKY130 open PDK, ORFS platform sky130hd	config.mk line 11	VERIFIED
Standard cells	sky130_fd_sc_hd (521 x sky130_fd_sc_hd__dfxtp_1)	synth_stat.txt line 219	VERIFIED
Liberty for synthesis, optimisation, ORFS final STA (TT only), power	sky130_fd_sc_hd__tt_025C_1v80 (1.8 V, 25 C, process 1.0; sha256 ec0e1067a35c8bf2...)	6_report.log line 6; 1_2_yosys.log line 213	VERIFIED
Liberty for informational STA	ss_100C_1v60, ff_n40C_1v95 from the master branch of efabless/skywater-pdk-libs-sky130_fd_sc_hd (16-hex sha256 prefixes only, no commit)	sta_corners_sep.txt lines 8-9	REPRODUCED 2026-10-06
PDK version	not recorded in any log; the TT liberty header has only revision : 1.0000000000. Needed: open_pdk/skywater-pdk commit per run	—	MISSING
Parasitics	OpenRCX, rcx_patterns.rules, one extraction corner; no RC corners	6_report.log lines 16-18	VERIFIED
Tool versions	OpenROAD "unknown"; Yosys 0.68+post (ORFS synthesis); KLayout 0.30.12	6_report.log line 1; 1_2_yosys.log line 1; 6_lvs.log line 1	VERIFIED
ORFS image	run used tag openroad/orfs:latest, no digest logged; requirements-tools.txt line 13 pins sha256:2e5bf6fe865e..., the only local image	—	ASSUMPTION

Item	Value	Source	Status
ORFS/OpenROAD commit, kepler-formal version	not recorded	—	MISSING

Supply voltages

No supply specification exists: SPEC.md, the RTL and the brief state no core or I/O supply voltage, tolerance, temperature range or power-up sequence (MISSING; needed: a SPEC section with VDD nominal/min/max and junction temperature range). The only evidence:

Item	Value	Source	Status
Library corner	TT, 1.80 V, 25 C. A characterisation corner, not a requirement	6_report.log line 6	VERIFIED
Supply nets in the layout	one pair. DEF PINS 217 = 215 signal + VDD (USE POWER) + VSS (USE GROUND), met5 strap pins; SPECIALNETS 2: VDD to VPB/VPWR, VSS to VNB/VGND	6_final.def.gz , decompressed lines 18346-18347, 18362, 19239-19240, 23252	VERIFIED
Static IR analysis supply	"Supply voltage : 1.80e+00 V". Not set in pd/sky130hd_sep/config.mk; it is the platform default PWR_NETS_VOLTAGES ?= VDD 1.8 (image platforms/sky130hd/config.mk line 145, read 2026-10-06)	6_report.log line 32	VERIFIED
Informational STA voltages	1.60 V (SS), 1.95 V (FF)	E-STA	REPRODUCED 2026-10-06
Power intent (UPF/CPF)	none; single domain in the layout. Needed: a SPEC statement that one domain is intended	—	MISSING

Interfaces

SPEC section 2 checked against the RTL ([orbit_demo.v](#) lines 26-51; [rtl_ports.json](#)) and the routed DEF pins ([pinout.csv](#)). Status of the table: REPRODUCED 2026-10-06 (Yosys 0.69+154 elaboration, parse_pins.py). All 18 ports match in name, direction and width.

Port	Dir	Width	Meaning (SPEC §2)	RTL line	DEF pins (edge)
clk	in	1	Clock	26	W
rst_n	in	1	Synchronous reset, active low	27	W
in_valid / in_ready	in / out	1 / 1	Input beat valid / accepted this cycle if in_valid	30 / 31	W / W
in_first / in_last	in	1 / 1	Beat starts a new sum / after this beat copy all sums to the buffer	32 / 33	S / W
in_a, in_b	in	32 each	Signed INT8 operands, lane i at [8*i +: 8]	34, 35	in_a S8 N10 W6 E8; in_b S9 N10 W6 E7
out_valid / out_ready	out / in	1 / 1	Buffer holds a presentable result / consumer takes it	38 / 39	W / W
out_data	out	128	Result, lane i at [32*i +: 32], signed INT32 (copy A)	40	W82 S24 N22
temp_valid	in	1	Thermal reading valid	43	E
temp_c	in	8	Signed whole degrees C (1)	44	E
clear_fault	in	1	Clear fault, zero lane storage, empty buffer	47	E
fault	out	1	Sticky mismatch flag, 1-cycle latency	48	W
therm_state	out	2	0 NORMAL, 1 THROTTLE, 2 STOP, 3 as STOP	49	E
therm_repair	out	1	Thermal copies disagree this cycle	50	E

Port	Dir	Width	Meaning (SPEC §2)	RTL line	DEF pins (edge)
shutdown_req	out	1	therm_state[1]	51	E

(1) Declared input wire [7:0] and reinterpreted as wire signed [7:0] t in [orbit_thermal_tmr.v](#) line 42, so comparisons are signed.

- 215 signal pins: 131 on met3 (E/W edges), 84 on met2 (N/S edges), minimum-size block-edge pins placed by `place_pins`; not a specified pinout, not bond pads. VDD/VSS are not in the SPEC table. Details: 08_pinout_packaging.
- Combinational input-to-output paths: `out_ready` -> `in_ready` (documented) and `clear_fault` -> `in_ready` (undocumented; see discrepancy register (09_review_notes)).
- I/O timing: 1.44 ns (0.2 x period) input and output delay "for the (unknown) outside world" ([constraint.sdc](#) lines 5-7, 20-27): ASSUMPTION. No `set_driving_cell`, `set_load`, `set_input_transition` or `set_clock_uncertainty` in [6_final.sdc](#): MISSING (needs an interface timing budget and clock jitter spec).
- Reset protocol (minimum `rst_n` low time; no on-chip synchronizer) and sensor interface timing (update rate, synchronisation to `clk`): MISSING from SPEC.

Target speed

No speed requirement exists for the demonstrator (MISSING; needed: target frequency and PVT range). The 7.2 ns period ([constraint.sdc](#) line 19) is the shortest period tried that closed at TT with the fences: 7.10 ns fails with setup WNS -0.094 ns ([period_exploration.md](#)). Conditions for all rows: sep layout, 7.2000 ns (138.9 MHz), 1.44 ns I/O delay, propagated clock, nominal OpenRCX SPEF at every corner, no OCV/derate or clock uncertainty, ORFS optimisation at TT only, OpenSTA in OpenROAD "unknown".

Result	Setup WNS / TNS (ns)	Hold WNS (ns)	Min period / fmax	Source	Status
ORFS final STA, TT 25 C 1.80 V	+0.031 / 0	+0.436	7.17 ns / 139.49 MHz	6_finish.rpt lines 5, 15, 20; 6_report.json lines 32-35	VERIFIED
Re-run TT tt_025C_1v80	+0.031 / 0.000	+0.436	7.17 ns / 139.49 MHz	sta_tt.log	REPRODUCED 2026-10-06
Re-run SS ss_100C_1v60	-6.131 / -2463.532, FAILS (516 endpoints)	+0.891	13.33 ns / 75.01 MHz	sta_ss_100C_1v60.log lines 6-11	REPRODUCED 2026-10-06
Re-run FF ff_n40C_1v95	+2.276 / 0.000	+0.281	4.92 ns / 203.07 MHz	sta_ff_n40C_1v95.log	REPRODUCED 2026-10-06

- TT worst setup path `in_a[27] -> g_lane[3].u_lane.u_res_a/q[29]$_SDFFE_PN0P_` includes the assumed 1.44 ns input delay; 139.49 MHz is from `report_clock_min_period -include_port_paths`. TT reg-to-reg worst slack is 1.346 ns ([supplement](#)).
- SS fails also reg-to-reg (-4.420 ns) and has 37 max-slew and 12 max-capacitance violating pins; TT and FF have none.
- Brief 200-800 MHz: TARGET for the concept chip only; 200 MHz is above the demonstrator's TT fmax.
- Throughput: no tool reports it. [docs/research/space-compute-competitors.md](#) line 25 "1.1 GOPS INT8 peak (4 MACs x 2 ops x 139 MHz)": CLAIM (unverified), a TT-only arithmetic bound.
- RC corners, other PVT corners, SDF-annotated simulation and multi-corner optimisation: MISSING.

Power budget

No power budget exists for the demonstrator (MISSING; needed: a target from the design owner). The brief's 40 W per chip is a concept-chip sizing ASSUMPTION and does not apply to `orbit_demo`.

Result	Value	Conditions / source	Status
Total power, routed sep layout	52.1092 mW: internal 26.3633 mW, switching 25.7459 mW, leakage 0.0279 uW	OpenSTA report_power (OpenROAD "unknown"), 7.2 ns, TT 25 C 1.80 V, OpenRCX SPEF, default activity; 6_report.json line 49, results.md lines 53-56	VERIFIED (tool estimate, not a measurement)
By group	Sequential 5.96 mW (11.4 %), Combinational 43.2 mW (83.0 %), Clock 2.90 mW (5.6 %), Macro 0, Pad 0	power_default_activity.txt ; 6_finish.rpt lines 549-560	VERIFIED
Activity assumed	the files state only "the tool's DEFAULT switching activity (no simulation activity annotated)". The ORFS scripts in the image call report_power with no set_power_activity, read_vcd or read_saif (grep of flow/scripts, 2026-10-06), so OpenSTA's built-in default applies; no log prints its toggle rate or static probability. Needed: VCD/SAIF from a representative workload read into OpenSTA	power_default_activity.txt	MISSING
Static IR drop, same activity	VDD 0.4539 mV, VSS 0.6275 mV worst (0.03 %)	PSM, 1.80 V; 6_report.log lines 27-49	VERIFIED
Power at SS/FF, dynamic IR, electromigration	not run	—	MISSING

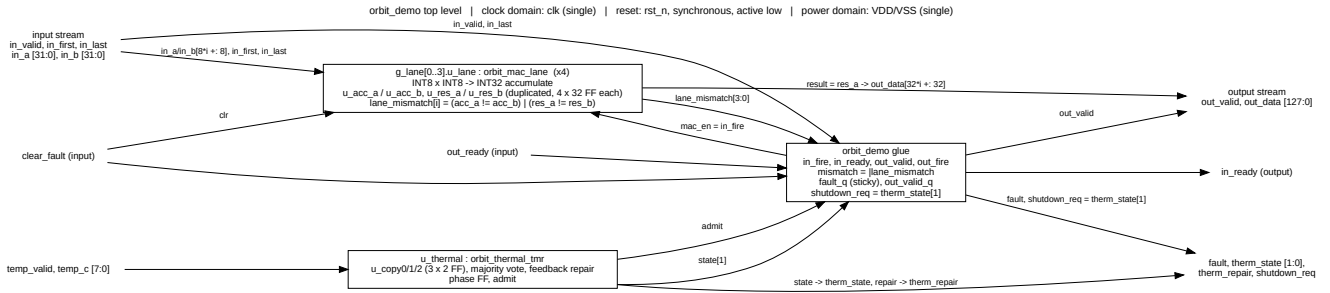
Area and cell counts

Conditions: ORFS sky130hd sep run 2026-09-29, TT liberty; ORFS synthesis by Yosys 0.68+post with abc -D 7.2.

Quantity	Value	Source	Status
Die	346.295 x 346.295 um = 119920 um ²	2_1_floorplan.log line 20; DEF DIEAREA	VERIFIED
Core	(2.300, 2.720)-(344.080, 342.720) um = 341.780 x 340.000 um = 116205.2 um ² ; 125 rows x 743 unithd sites	2_1_floorplan.log lines 19-22	VERIFIED
Utilisation	CORE_UTILIZATION 40 % (config); 0.403 at floorplan; 0.536 final (cell area excl. fill / core)	config.mk line 35; 2_1_floorplan.log line 24; 6_report.json line 64	VERIFIED
Cell area	62299.8 um ² excl. fill incl. tap; 60392.97 um ² excl. fill and tap	6_report.log line 67; results.md lines 42-43	VERIFIED
Instances	18175: fill 10381, tap 1524, antenna 20, clock buffer 88, clock inverter 8, timing-repair buffer 1201, inverter 171, sequential 521, multi-input combinational 4261. 6270 excl. fill/tap; 6_final.v has 6250 leaf cells	6_report.log lines 51-61; storage_audit.txt line 3	VERIFIED
Macros, pad cells	0, 0	6_report.json	VERIFIED
Synthesised netlist	5130 cells, 46866.1984 um ² (sequential 10430.0032 um ² , 22.25 %); 19 kept orbit_keep_reg instances	synth_stat.txt lines 72, 268-269	VERIFIED
Generic synthesis	4899 cells = 4373 logic + 521 flip-flops + 5 \$scopeinfo (Yosys 0.69+154, no area)	synth/synth_counts.txt	VERIFIED

Only the flip-flop count (521) is the same across the generic, synthesised and routed netlists; the other counts are not comparable. Layout views: 06_physical_design.

Top-level block diagram



Drawn from rtl/*.v at 26d71a8 (Graphviz source [orbit_demo_top_level.dot](#); [PDF](#)). Instance names are the RTL names. One clock domain (clk), one synchronous active-low reset (rst_n), one supply pair (VDD/VSS). Detailed diagram: [orbit_demo_block_diagram.svg](#).